

More on Specification and Data Issues

Intermediate Econometrics

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Wooldridge, Chapter 9

Outline

Functional Form Misspecification (§9-1)

Proxy Variables for Unobserved Variables (§9-2)

Models with Random Slopes (§9-3)

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Missing Data, Nonrandom Samples, Outliers (§9-5)

Least Absolute Deviations Estimation (§9-6)

Functional Form Misspecification (§9-1)

Where We Are

Chapter 8 dealt with one Gauss–Markov failure: **heteroskedasticity**. This was a mild problem: OLS remained **unbiased and consistent**, only inference needed adjustment.

Chapter 9 returns to a much more serious problem: **correlation between u and one or more explanatory variables** ($\text{Cov}(x_j, u) \neq 0$). This causes **bias and inconsistency** in all OLS estimators.

Three main reasons x_j can be endogenous (this chapter):

1. **Functional form misspecification** (§9-1) — omitting nonlinear terms
2. **Omitted variables** (§9-2) — addressed via *proxy variables*
3. **Measurement error** (§9-4) — biases of known sign

Plus three **data problems**:

- ▶ Missing data (§9-5a)
- ▶ Nonrandom sampling (§9-5b)
- ▶ Outliers and the LAD alternative (§9-5c, §9-6)

Functional Form Misspecification: Motivation & Setup

Motivation. Even with every relevant variable collected, OLS can be biased if y is related to the x_j through the **wrong functional form** — a missing quadratic, log, or interaction. We *have* the data; only the specification is wrong.

Setup. If the true model carries a nonlinear term we drop, that term is absorbed into the error and induces $\text{Cov}(x_j, u) \neq 0$:

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + \underbrace{\gamma x_1^2 + \cdots}_{\text{dropped} \rightarrow u}$$

Plan: (i) what misspecification *is*, (ii) detect it — F test on quadratics and the **RESET** test, (iii) choose between **nonnested** models.

Functional Form Misspecification: Definition

Functional Form Misspecification

A multiple regression suffers from **functional form misspecification** when it does not properly account for the relationship between the dependent variable and the observed explanatory variables.

Example: A Wage Equation

True model:

$$\log(\text{wage}) = \beta_0 + \beta_1 \text{educ} + \beta_2 \text{exper} + \beta_3 \text{exper}^2 + u.$$

If we estimate

$$\log(\text{wage}) = \beta_0 + \beta_1 \text{educ} + \beta_2 \text{exper} + u$$

(omitting exper^2), we have functional form misspecification.

Consequence: Generally biased estimates of $\beta_0, \beta_1, \beta_2$. The **return to experience** is no longer β_2 but $\beta_2 + 2\beta_3 \text{exper}$, so a single number cannot capture it.

Another Example: Missing Interactions

Consider the wage equation:

$$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 exper^2 + \beta_4 female + \beta_5 (female \cdot educ) + u.$$

If we **omit the interaction** $female \cdot educ$:

- ▶ We **misspecify** the functional form.
- ▶ Estimators of *all* other parameters are biased.
- ▶ The "return to education" is unclear because it depends on **gender**.

Key insight: Functional form misspecification = the *wrong functional form* of *observed* variables. We **have** the data; we are just not using it correctly.

Contrast this with **omitted variables** (§9-2), where data on a key variable *cannot be collected*.

Detecting Misspecification: The F Test for Quadratics

We already have a tool: the F test for joint exclusion restrictions.

Procedure:

1. Add **quadratic terms** for any significant explanatory variable.
2. Perform a joint F test of significance.
3. If significant: keep the quadratics (at the cost of complicating interpretation).

Symptomatic warning: Significant quadratic terms can also indicate other functional form problems — e.g., using the **level** of a variable when the **logarithm** would be more appropriate, or vice versa.

Adding quadratics + using logarithms covers many nonlinear relationships in economics.

Example 9.1: Economic Model of Crime (CRIME1)

Dependent variable: *narr86* (number of arrests in 1986). $n = 2,725$.

Variable	(1) Linear	(2) With quadratics
<i>pcnv</i>	-0.133 (0.040)	0.553 (0.154)
<i>pcnv</i> ²	—	-0.730 (0.156)
<i>ptime86</i>	-0.041 (0.009)	0.287 (0.094)
<i>ptime86</i> ²	—	-0.0296 (0.0039)
<i>inc86</i>	-0.0015 (0.0003)	-0.0034 (0.0008)
<i>inc86</i> ²	—	-0.000007 (0.000003)
<i>R</i> ²	0.0723	0.1035

Joint F test on the three squared terms: $F = 31.37$ ($df = 3, 2713$), p -value ≈ 0 .
The squared terms are **jointly very significant**.

Interpretation: The initial linear model overlooked important nonlinearities. But interpreting the quadratics is harder: e.g., *pcnv* has a positive deterrent effect only up to $pcnv = 0.365$, then negative.

The RESET Test

Adding quadratics is one approach but uses up many degrees of freedom. A more general test: **Ramsey's RESET (1969)**.

RESET Idea

Idea: if the original model

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + u$$

satisfies MLR.4, then **no nonlinear function of the x_j** should be significant when added.

RESET adds polynomials in the OLS fitted values $\hat{y}$ to detect general forms of misspecification. Squared and cubed terms are commonly used.

Why $\hat{y}$ instead of all the x_j 's? $\hat{y}$ summarizes *all* the regressors in one variable, saving degrees of freedom.

The RESET Statistic

Let $\hat{y}$ be OLS fitted values from the original model. Estimate the **augmented regression**:

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + \delta_1 \hat{y}^2 + \delta_2 \hat{y}^3 + \text{error}.$$

RESET test:

$$H_0 : \delta_1 = 0, \delta_2 = 0 \quad (\text{model correctly specified})$$

$$F \sim F_{2, n-k-3} \quad (\text{approximately, in large samples}).$$

A **significant** F statistic suggests functional form misspecification.

Notes:

- ▶ An LM version is also available (chi-square distribution, 2 df).
- ▶ Both versions are easily made robust to heteroskedasticity (Ch. 8 methods).

Example 9.2: Housing Price Equation (HPRICE1, $n = 88$)

Model 1 (levels):

$$price = \beta_0 + \beta_1 lotsize + \beta_2 sqrft + \beta_3 bdrms + u.$$

Model 2 (log-log):

$$\log(price) = \beta_0 + \beta_1 \log(lotsize) + \beta_2 \log(sqrft) + \beta_3 bdrms + u.$$

	RESET F statistic	p -value
Model 1 (levels)	4.67	0.012
Model 2 (log-log)	2.56	0.084

Verdict:

- ▶ Reject Model 1 at the 5% level $\Rightarrow$ misspecification.
- ▶ Do not reject Model 2 at the 5% level (would reject at 10%).
- ▶ **The log-log model is preferred** based on RESET.

Cautions about RESET

What RESET can do:

- ▶ Detect general functional form misspecification (conditional mean).
- ▶ Compare two non-nested specifications informally.

What RESET cannot do:

- ▶ Detect **omitted variables** when those variables have linear conditional expectations in the included regressors.
- ▶ Detect **heteroskedasticity** (RESET is for the conditional *mean*).

Bottom line: RESET is a functional form test, **nothing more**.

Practical limitation: If RESET rejects, it does *not* tell you what to do next. You must investigate further (try logs, add interactions, use quadratics, etc.).

Tests against Nonnested Alternatives (§9-1b)

Some specifications are not nested in each other. Consider:

$$\text{Model A: } y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u \quad (9.6)$$

$$\text{Model B: } y = \beta_0 + \beta_1 \log(x_1) + \beta_2 \log(x_2) + u \quad (9.7)$$

Nonnested Models

Two models are **nonnested** if neither can be obtained from the other by imposing parameter restrictions. (Here: levels vs. logs.)

Cannot use a standard F test. Two approaches:

1. **Comprehensive (Mizon–Richard 1986):** construct a model nesting both, test restrictions.
2. **Davidson–MacKinnon (1981):** an auxiliary t test (next slide).

The Davidson–MacKinnon Test

Idea: If Model A is correct, then fitted values from the *other* model (Model B) should be **insignificant** when added to Model A.

Procedure (testing A against B):

1. Estimate Model B by OLS, save fitted values $\check{y}$.
2. Run the auxiliary regression:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \theta_1 \check{y} + \text{error}.$$

3. **t -statistic on θ_1** is the Davidson–MacKinnon statistic.

Decision: A significant t -statistic (two-sided alternative) is **evidence against Model A**.

To test Model B against Model A: swap roles. Save fitted values $\hat{y}$ from A, add to B as a regressor.

Caveats of Nonnested Testing

Three problems with nonnested tests:

1. **No clear winner needed:** Both models could be rejected, or neither rejected. If both rejected: more work needed. If neither: report both, use adjusted R^2 or theoretical considerations.
2. **Rejecting A does not validate B:** Model A can be rejected for any number of reasons, not just the one B captures.
3. **Different dependent variables:** Comparing y vs. $\log(y)$ requires additional adjustments. Goodness-of-fit comparisons need care (Wooldridge 1994a).

Proxy Variables for Unobserved Variables (§9-2)

Proxy Variables: Motivation & Setup

Motivation. The most damaging specification error is a **missing determinant we cannot observe** — e.g. *ability* in a wage equation. If it correlates with a regressor, OLS is biased and inconsistent (omitted-variable bias). Can we do anything *without* the variable itself?

Setup. We cannot drop the unobservable (that *is* the bias) and we cannot measure it. The compromise: substitute an observed **proxy** that is correlated with it. **Plan:** the proxy idea, the plug-in regression, the conditions under which it removes the bias, and using a **lagged dependent variable** as a proxy.

Omitted Variables: A Bigger Problem

Functional form misspecification is mild: we have all the data, just need the right form.

Omitted variables are harder: a key variable is unobserved, often because data *cannot be collected*.

Wage equation example:

$$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 abil + u$$

where *abil* = ability — **unobservable**. If *educ* is correlated with *abil*, putting *abil* in the error causes **omitted variable bias**.

Question: Can we mitigate this bias if we cannot directly observe *abil*?

Answer: Sometimes — using a **proxy variable**.

The Proxy Variable Idea

Proxy Variable

A **proxy variable** is something *related to* the unobserved variable, used as a **substitute** in the regression.

Example: **IQ score** as a proxy for *ability*.

- ▶ Does *not* require $\text{IQ} = \text{ability}$.
- ▶ Requires IQ to be **correlated** with ability.

Setup: Consider the model with three explanatory variables (two observed):

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3^* + u, \quad (9.10)$$

where x_3^* is **unobserved** but we have a proxy x_3 :

$$x_3^* = \delta_0 + \delta_3 x_3 + v_3. \quad (9.11)$$

The intercept δ_0 allows x_3^* and x_3 to have different scales (e.g., ability has no natural scale, but IQ has mean 100).

The Plug-in Solution

Plug-in Solution to Omitted Variables

Pretend that x_3 and x_3^* are the same. Run:

$$y \text{ on } x_1, x_2, x_3. \quad (9.12)$$

Use x_3 as a **stand-in** for the unobservable x_3^* .

When does this work? Two assumptions are sufficient:

1. u **uncorrelated with** x_1, x_2, x_3^*, x_3 : The first three are standard. The new requirement: u uncorrelated with x_3 . Holds if x_3 is **irrelevant** once x_3^* is controlled for.
2. v_3 **uncorrelated with** x_1, x_2, x_3 : Means x_3 is a **good proxy** for x_3^* :

$$\mathbb{E}(x_3^* \mid x_1, x_2, x_3) = \mathbb{E}(x_3^* \mid x_3) = \delta_0 + \delta_3 x_3. \quad (9.13)$$

Why the Plug-in Solution Works

Plug the proxy equation $\mathbf{x}_3^* = \delta_0 + \delta_3 \mathbf{x}_3 + \mathbf{v}_3$ into the model:

$$y = (\beta_0 + \beta_3 \delta_0) + \beta_1 \mathbf{x}_1 + \beta_2 \mathbf{x}_2 + \beta_3 \delta_3 \mathbf{x}_3 + (\mathbf{u} + \beta_3 \mathbf{v}_3).$$

Define the composite error $\mathbf{e} = \mathbf{u} + \beta_3 \mathbf{v}_3$. Then:

$$y = \alpha_0 + \beta_1 \mathbf{x}_1 + \beta_2 \mathbf{x}_2 + \alpha_3 \mathbf{x}_3 + \mathbf{e},$$

where $\alpha_0 = \beta_0 + \beta_3 \delta_0$ and $\alpha_3 = \beta_3 \delta_3$.

What we get:

- ▶ **Unbiased estimates of β_1 and β_2** (the parameters of interest).
- ▶ **Unbiased estimate of $\alpha_3 = \beta_3 \delta_3$** (interpretable as the partial effect of the proxy on y).
- ▶ **Cannot recover β_3 alone** (we do not observe δ_3).

Often α_3 is more interesting anyway: e.g., the wage premium per IQ point is more directly useful than the wage premium per "unit of ability."

Example 9.3: IQ as a Proxy for Ability (WAGE2)

Data: WAGE2 (Blackburn–Neumark 1992), $n = 935$ men. Includes monthly earnings, education, demographics, and IQ scores from 1980.

	(1) No IQ	(2) With IQ	(3) With IQ + interaction
<i>educ</i>	0.065 (0.006)	0.054 (0.007)	0.053 (0.007)
<i>exper</i>	0.014 (0.003)	0.014 (0.003)	0.014 (0.003)
<i>tenure</i>	0.012 (0.002)	0.011 (0.002)	0.011 (0.002)
<i>IQ</i>	—	0.0036 (0.0010)	0.0036 (0.0010)
$(educ - \overline{educ})(IQ - \overline{IQ})$	—	—	0.00034 (0.00038)
R^2	0.253	0.263	0.263

Key finding:

- ▶ Without IQ: estimated return to education = **6.5%**.
- ▶ With IQ as proxy: return falls to **5.4%**.
- ▶ The drop (1.1 percentage points) corresponds to omitted ability bias.
- ▶ IQ premium: 3.6% per 10 IQ points (since SD of IQ ≈ 15 , a one-SD increase $\Rightarrow$ 5.4% higher earnings).

The interaction ($educ \cdot IQ$) in column (3) is insignificant ($t \approx 0.89$): no convincing evidence of an interactive effect.

When the Proxy Is Imperfect

Suppose the proxy depends on the other regressors:

$$\mathbf{x}_3^* = \delta_0 + \delta_1 \mathbf{x}_1 + \delta_2 \mathbf{x}_2 + \delta_3 \mathbf{x}_3 + \mathbf{v}_3. \quad (9.14)$$

Then plugging in:

$$y = (\beta_0 + \beta_3 \delta_0) + (\beta_1 + \beta_3 \delta_1) \mathbf{x}_1 + (\beta_2 + \beta_3 \delta_2) \mathbf{x}_2 + \beta_3 \delta_3 \mathbf{x}_3 + (\mathbf{u} + \beta_3 \mathbf{v}_3).$$

Bias formula:

$$\text{plim}(\hat{\beta}_1) = \beta_1 + \beta_3 \delta_1, \quad \text{plim}(\hat{\beta}_2) = \beta_2 + \beta_3 \delta_2.$$

Bias remains, but **should be smaller** than ignoring the omitted variable entirely.

Wage example: If $\delta_1 > 0$ (more education $\Rightarrow$ higher ability) and $\beta_3 > 0$ (ability $\Rightarrow$ higher wage), then $\hat{\beta}_1$ remains upward biased *even with IQ as proxy*. But less than without IQ.

Lagged Dependent Variables as Proxies (§9-2a)

Sometimes we do not even have a candidate proxy. A useful alternative: **the lagged dependent variable**.

Example 9.4: City Crime Rates

$$crime = \beta_0 + \beta_1 unem + \beta_2 expend + \beta_3 crime_{-1} + u. \quad (9.16)$$

- ▶ $crime_{-1}$ = crime rate from an earlier year.
- ▶ $\beta_3 > 0$ expected: crime has **inertia**.
- ▶ Including $crime_{-1}$ controls for **historical unobservables** that drive both past and current crime.

Reading β_2 : The effect on crime of an extra dollar of law enforcement, **holding constant prior crime rate**. This is a much cleaner causal interpretation than a pure cross-sectional regression.

Crime Rate Estimates (CRIME2, $n = 46$ cities)

Variable	(1) No lag	(2) With lagged crime
$unem_{87}$	-0.029 (0.032)	0.009 (0.020)
$\log(lawexp_{87})$	0.203 (0.173)	-0.140 (0.109)
$\log(crmrte_{82})$	—	1.194 (0.132)
<i>intercept</i>	3.34 (1.25)	0.076 (0.821)
R^2	0.057	0.680

Findings:

- ▶ **Without** lagged crime: counterintuitive — more law enforcement spending seems to *increase* crime ($t \approx 1.17$).
- ▶ **With** lagged crime: elasticity of crime w.r.t. enforcement becomes -0.14 ($t \approx -1.28$).
- ▶ Lagged crime is highly significant (elasticity ≈ 1): cities with high past crime tend to have high current crime.

Why the original sign was wrong: Cities with high recent crime spend more on law enforcement, creating reverse correlation. Lagged crime "soaks up" this confounding.

Why Was Unemployment Negative Without the Lag?

Theory predicts $\beta_{unem} \geq 0$ (joblessness raises crime), yet column (1) reports -0.029 . Two points resolve the puzzle.

1. It is not significant. $t = -0.029/0.032 \approx -0.9$ — the negative estimate is **statistically indistinguishable from zero**. The “wrong sign” is sampling noise, not a finding: never over-read a single insignificant coefficient.

2. Omitted-variable bias. Column (1) leaves out **persistent city traits** (demographics, socioeconomic conditions, reporting practices) that drive crime; $\log(crmrte_{82})$ proxies for them. With δ = effect of those traits on crime and γ = their partial association with $unem$,

$$\text{plim } \hat{\beta}_{unem}^{(1)} = \beta_{unem} + \delta \gamma.$$

The coefficient *rises* ($-0.029 \rightarrow +0.009$) once the lag is added, so the bias $\delta\gamma$ was **negative**. Since $\delta > 0$ (past crime enters at $+1.19$), we infer $\gamma < 0$: in this sample, higher-baseline-crime cities had *lower* 1987 unemployment.

After the lag: the confounding is absorbed and $\hat{\beta}_{unem}$ returns to its expected non-negative sign ($+0.009$) — though still insignificant at $n = 46$.

A Different Slant on Multiple Regression (§9-2b)

Alternative interpretation of regression with proxies:

Instead of seeing IQ as a proxy for the latent concept "ability," interpret the wage equation more modestly:

"Among workers with the same IQ, experience, etc., what is the expected difference in log wage from one more year of education?"

Predictive perspective:

- ▶ Sometimes the goal is **prediction**, not causal inference.
- ▶ Then "omitted variable bias" loses its meaning.
- ▶ Focus instead on building a good predictive model with available regressors.
- ▶ Do not include variables **unobserved at the time of prediction**.

Example: A college admissions office uses high school GPA, SAT, etc. to predict college GPA. They are not concerned about omitted variable bias from omitting "motivation."

Potential Outcomes and Proxy Variables (§9-2c)

The proxy-variable idea links directly to the **potential outcomes** framework (Ch. 2–7). For a binary treatment $w \in \{0, 1\}$ with potential outcomes $y(0), y(1)$:

$$y(0) = \mu_0 + v(0), \quad y(1) = \mu_1 + v(1), \quad \tau_{ate} = \mu_1 - \mu_0.$$

Selection problem: participation w may be correlated with the unobservables $v(0), v(1)$. **Unconfoundedness (ignorability):** conditional on covariates $\mathbf{x}$, w is independent of $(y(0), y(1))$ — which is *exactly the proxy-variable condition*: $\mathbf{x}$ proxies for the unobserved factors driving both the outcome and the treatment decision.

Estimation. With linear forms $\mathbb{E}[v(g) \mid w, \mathbf{x}] = \mathbb{E}[v(g) \mid \mathbf{x}] = (\mathbf{x} - \boldsymbol{\eta})\boldsymbol{\beta}_g$, $\boldsymbol{\eta} = \mathbb{E}(\mathbf{x})$, the ATE comes from one interacted regression over the *entire* sample,

$$y_i \text{ on } w_i, \mathbf{x}_i, w_i \cdot (\mathbf{x}_i - \bar{\mathbf{x}}),$$

where the coefficient on w_i is $\hat{\tau}_{ate}$ (derived next, §7-6).

The Parameter of Interest and the ATE Regression (§7-6)

Setup. Each unit has potential outcomes $y(1)$ (if treated) and $y(0)$ (if untreated); with treatment $w \in \{0, 1\}$ we observe only one, $y = (1 - w)y(0) + wy(1)$, where $y(g) = \mu_g + v(g)$ and $\mu_g = \mathbb{E}[y(g)]$.

Parameter of interest — the average treatment effect (ATE):

$$\tau = \mathbb{E}[y(1) - y(0)] = \mu_1 - \mu_0.$$

The *unit-level* effect $y_i(1) - y_i(0)$ is never observed, so we target its average.

Derivation. Take $\mathbb{E}(\cdot \mid w, \mathbf{x})$; **unconfoundedness** drops w from each conditional mean:

$$\begin{aligned}\mathbb{E}(y \mid w, \mathbf{x}) &= \mathbb{E}[y(0) \mid \mathbf{x}] + w \{ \mathbb{E}[y(1) \mid \mathbf{x}] - \mathbb{E}[y(0) \mid \mathbf{x}] \} \\ &= \mu_0 + (\mathbf{x} - \boldsymbol{\eta})\boldsymbol{\beta}_0 + \underbrace{(\mu_1 - \mu_0)}_{\tau} w + (\boldsymbol{\beta}_1 - \boldsymbol{\beta}_0) w(\mathbf{x} - \boldsymbol{\eta}).\end{aligned}$$

$\Rightarrow$ regress y on $w, \mathbf{x}, w(\mathbf{x} - \bar{\mathbf{x}})$ ($\boldsymbol{\eta} = \mathbb{E}(\mathbf{x})$ replaced by $\bar{\mathbf{x}}$); the coefficient on w estimates τ .

Why That Form? Interaction and Centering (§7-6)

Why the interaction $w(\mathbf{x} - \bar{\mathbf{x}})$? It lets the treated and control groups have *different slopes* ($\beta_1 \neq \beta_0$), so the effect may vary with covariates:

$$\tau(\mathbf{x}) = \tau + (\mathbf{x} - \boldsymbol{\eta})(\beta_1 - \beta_0).$$

Dropping it forces a *constant* effect $\tau(\mathbf{x}) \equiv \tau$.

Why center at $\bar{\mathbf{x}}$? The ATE is the *average* of $\tau(\mathbf{x})$, and $\mathbb{E}(\mathbf{x} - \boldsymbol{\eta}) = 0$, so the coefficient on w equals τ **only when the interaction is demeaned**. With a raw $w\mathbf{x}$ term, that coefficient would instead be the effect at $\mathbf{x} = \mathbf{0}$ — a (usually fictional) unit, not the population average.

Models with Random Slopes (§9-3)

Models with Random Slopes: Motivation & Setup

Motivation. Every model so far forces *one* slope β_j on the whole population. But the return to education, or the effect of a price change, plausibly **differs across individuals** by unobserved traits. What is OLS estimating when slopes are heterogeneous?

Setup. Allow each unit its own intercept a_i and slope b_i , and define the population averages $\alpha = \mathbb{E}(a_i)$ and $\beta = \mathbb{E}(b_i)$, so that β is the **average partial effect (APE)**. **Plan:** the conditions under which OLS still recovers α, β , the heteroskedasticity it induces, and when it breaks down.

The Random Slopes Model

Until now, we have assumed all individuals share the same slope β_j . What if the slope varies across the population by **unobservable** characteristics?

Random Coefficient (Random Slope) Model

For unit i in the population:

$$y_i = a_i + b_i x_i, \quad (9.17)$$

where:

- ▶ a_i = unit-specific **intercept**
- ▶ b_i = unit-specific **slope**
- ▶ Both vary across the population

Example: Wage equation with b_i = individual return to education. Different people benefit differently from another year of schooling depending on **unobserved ability, motivation**, etc.

Define $\alpha = \mathbb{E}(a_i)$ and $\beta = \mathbb{E}(b_i)$ — the **average intercept** and **average partial effect (APE)**.

When OLS Still Works

Write $a_i = \alpha + c_i$, $b_i = \beta + d_i$ with $\mathbb{E}(c_i) = 0$, $\mathbb{E}(d_i) = 0$. Substituting into (9.17):

$$y_i = \alpha + \beta x_i + (c_i + d_i x_i) = \alpha + \beta x_i + u_i, \quad (9.18)$$

where the new error is $u_i = c_i + d_i x_i$.

Sufficient conditions for OLS to estimate α and β unbiasedly:

$$\mathbb{E}(a_i | x_i) = \mathbb{E}(a_i), \quad \mathbb{E}(b_i | x_i) = \mathbb{E}(b_i). \quad (9.19)$$

That is: a_i and b_i are **mean independent** of x_i .

Then OLS recovers the **population averages** of the unit-specific slopes.

Inevitable side effect: The error $u_i = c_i + d_i x_i$ is **heteroskedastic**:

$$\text{Var}(u_i | x_i) = \sigma_c^2 + \sigma_d^2 x_i^2. \quad (9.20)$$

Use **heteroskedasticity-robust standard errors** (Ch. 8).

Random Slopes: Practical Takeaway

Key result: If we just want to estimate the **average partial effect** of x on y (the APE), and a_i, b_i are independent of x_i , then **OLS works**.

- ▶ Use OLS as before.
- ▶ Always use heteroskedasticity-robust standard errors.
- ▶ Interpretation changes: $\hat{\beta}$ is the **average** return, not the return for any specific person.

When OLS fails: If b_i is correlated with x_i (e.g., people who benefit more from education *select into* more education), OLS is biased. Solution: instrumental variables (Ch. 15).

Connection to Ch. 7–8: Random slopes provide **another justification** for using interaction terms or heteroskedasticity-robust SEs in standard regression analysis.

Properties of OLS under Measurement Error (§9-4)

Measurement Error: Motivation & Setup

Motivation. We often record a variable *with error*: reported vs true income, average vs marginal tax rate, self-reported vs actual hours. Does a mismeasured variable bias OLS — and if so, in **which direction**?

Setup. Write each observed value as truth plus noise, e.g. $x_1 = x_1^* + e_1$, and trace how the noise e_1 propagates into $\hat{\beta}$. **Plan:** (i) measurement error in the *dependent* variable (usually harmless), then (ii) error in an *explanatory* variable under the **classical errors-in-variables** assumption $\Rightarrow$ **attenuation bias** toward zero.

Measurement Error: The Issue

Sometimes we cannot collect data on the variable that truly affects economic behavior.

Example: Marginal Tax Rate

A family chooses charitable contributions based on their **marginal income tax rate**. But the marginal rate depends on income brackets and is hard to compute. Researchers often use **average tax rate** as a substitute — measurement error.

Measurement Error vs. Proxy Variables

Both involve a substitute for an unobserved variable, but:

- ▶ **Proxy:** the unobserved variable is *vague* (e.g., "ability"); the proxy (e.g., IQ) is just *associated* with it.
- ▶ **Measurement error:** the unobserved variable has a *well-defined, quantitative meaning*; we just cannot measure it precisely.

This section: **when does measurement error cause problems for OLS?**

Measurement Error in the Dependent Variable (§9-4a)

Let y^* be the true variable, y be the observed (measured) variable. The **measurement error** is

$$e_0 = y - y^*, \quad (9.24)$$

so $y = y^* + e_0$. Substituting into the true model:

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + (u + e_0). \quad (9.25)$$

Standard assumption: e_0 is **independent** of the x_j 's.

Under this assumption:

- ▶ $\mathbb{E}(e_0) = 0$ (or absorbed in intercept).
- ▶ OLS is **unbiased and consistent**.
- ▶ Inference (t , F , LM tests) is valid.
- ▶ **Only cost:** larger error variance $\Rightarrow$ larger SEs.

Bottom line: Random measurement error in y is innocuous — it only inflates standard errors. Cannot fix without better data.

Example 9.5: Savings Function

Consider a savings function:

$$sav^* = \beta_0 + \beta_1 inc + \beta_2 size + \beta_3 educ + \beta_4 age + u.$$

- ▶ Observed savings sav may differ from true sav^* due to reporting error.
- ▶ If measurement error is uncorrelated with inc , $size$, $educ$, age — OLS is fine.
- ▶ **But** families with higher income or more education may report savings *more accurately* — measurement error correlated with regressors $\Rightarrow$ bias.

Logarithmic case: If the dependent variable is $\log(y^*)$:

$$\log(y) = \log(y^*) + e_0,$$

which corresponds to a **multiplicative** measurement error in y : $y = y^* a_0$ with $e_0 = \log(a_0)$. Common in firm-level data, where reported figures may be a multiplicative fraction of true values.

Measurement Error in an Explanatory Variable (§9-4b)

Now the **important** case. Let x_1^* be true, x_1 observed:

$$e_1 = x_1 - x_1^*. \quad (9.28)$$

Standard assumptions:

- ▶ $\mathbb{E}(e_1) = 0$.
- ▶ u uncorrelated with x_1^* and x_1 (i.e., once x_1^* is controlled for, x_1 has no separate effect on y).

Two Polar Cases

1. e_1 uncorrelated with the **observed** x_1 :

$$\text{Cov}(x_1, e_1) = 0. \quad (9.29)$$

Like proxy variable case. OLS is consistent (with larger variance).

2. **Classical Errors-in-Variables (CEV):**

$$\text{Cov}(x_1^*, e_1) = 0. \quad (9.30)$$

The error is uncorrelated with the **true** variable. **Standard textbook assumption.**

The Attenuation Bias under CEV

Under CEV: $x_1 = x_1^* + e_1$ with $\text{Cov}(x_1^*, e_1) = 0$. Then:

$$\text{Cov}(x_1, e_1) = \text{Cov}(x_1^* + e_1, e_1) = \sigma_{e_1}^2.$$

Substituting $x_1^* = x_1 - e_1$ into $y = \beta_0 + \beta_1 x_1^* + u$:

$$y = \beta_0 + \beta_1 x_1 + (u - \beta_1 e_1). \quad (9.32)$$

The composite error $u - \beta_1 e_1$ is **correlated with** x_1 . Hence OLS is biased.

The bias formula:

$$\text{plim}(\hat{\beta}_1) = \beta_1 \cdot \frac{\sigma_{x_1^*}^2}{\sigma_{x_1^*}^2 + \sigma_{e_1}^2}. \quad (9.33)$$

The factor $\frac{\sigma_{x_1^*}^2}{\sigma_{x_1^*}^2 + \sigma_{e_1}^2}$ is between 0 and 1.

Conclusion: $\hat{\beta}_1$ is **always closer to zero** than β_1 . This is the **attenuation bias**.

How Big Is the Attenuation?

Define the **signal-to-noise ratio** $r = \sigma_{x_1^*}^2 / \sigma_{e_1}^2$.

$$\text{plim}(\hat{\beta}_1) = \beta_1 \cdot \frac{r}{r+1}.$$

Signal-to-noise r	Attenuation factor	Bias if $\beta_1 = 1$
$r = 1$ (50% noise)	0.50	$\hat{\beta}_1 \rightarrow 0.50$
$r = 4$ (20% noise)	0.80	$\hat{\beta}_1 \rightarrow 0.80$
$r = 9$ (10% noise)	0.90	$\hat{\beta}_1 \rightarrow 0.90$
$r = 99$ (1% noise)	0.99	$\hat{\beta}_1 \rightarrow 0.99$

Implications:

- ▶ Sign of $\hat{\beta}_1$ is preserved (only magnitude shrinks).
- ▶ If $\beta_1 > 0$, $\hat{\beta}_1$ **understates** the true effect.
- ▶ Hypothesis test of $H_0 : \beta_1 = 0$ has **less power**.
- ▶ More variation in x_1^* relative to $e_1 \Rightarrow$ less bias.

Multiple Regression with CEV

Suppose only x_1^* is mismeasured:

$$y = \beta_0 + \beta_1 x_1^* + \beta_2 x_2 + \beta_3 x_3 + u, \quad (9.34)$$

with x_2, x_3 measured perfectly. Under CEV:

Bias on the mismeasured variable:

$$\text{plim}(\hat{\beta}_1) = \beta_1 \cdot \frac{\sigma_{r_1^*}^2}{\sigma_{r_1^*}^2 + \sigma_{e_1}^2}, \quad (9.36)$$

where r_1^* is the **population residual** from regressing x_1^* on x_2, x_3 .

Bias on the other coefficients: Generally, $\hat{\beta}_2$ and $\hat{\beta}_3$ are **also inconsistent** (due to indirect correlation). Sizes and signs of these biases are hard to predict.

Bottom line: Measurement error in *one* regressor contaminates *all* OLS estimates.

Example 9.7: GPA Equation with Measurement Error

Suppose true model:

$$colGPA = \beta_0 + \beta_1 faminc^* + \beta_2 hsGPA + \beta_3 SAT + u,$$

where $faminc^*$ = actual annual family income (mismeasured if students self-report).

Observed: $faminc = faminc^* + e_1$.

If CEV holds:

- ▶ OLS estimate of β_1 is **biased toward zero**.
- ▶ Reduced power for $H_0 : \beta_1 = 0$.
- ▶ Practical fear: cannot reject "family income has no effect on college GPA" even if the true effect is nontrivial.

Solution preview: If a second *noisy measurement* of family income is available (or another instrument), **instrumental variables** (Ch. 15) can correct the attenuation bias.

Missing Data, Nonrandom Samples, Outliers (§9-5)

Data Problems: Motivation & Setup

Motivation. Real samples are imperfect: observations have **gaps**, the sample may be drawn **nonrandomly**, and a few **outliers** can dominate the fit. Each can threaten random sampling (MLR.2) — but not all of them bias OLS.

Setup. For every data problem the decisive question is the same: *is an observation's inclusion or exclusion related to the error u ?*

- ▶ **Missing data** (§9-5a): MCAR / MAR / NMAR; complete-case vs. the missing-indicator method.
- ▶ **Nonrandom samples** (§9-5b): *exogenous* (harmless) vs. *endogenous* (biasing) selection.
- ▶ **Outliers** (§9-5c): influential observations — which motivate **LAD** in §9-6.

Missing Data (§9-5a)

Three Types of Missingness

1. **Missing Completely At Random (MCAR):** the probability of missingness is *independent* of all observed and unobserved factors affecting y .
2. **Missing At Random (MAR):** probability of missingness can depend on observed regressors but not on unobservables.
3. **Not Missing At Random (NMAR):** missingness depends on the unobserved value itself.

If MCAR holds: dropping observations with missing data (the **complete cases estimator**) is OLS-consistent. Just a smaller sample.

Example: If 196 of 1,388 observations in BWGHT are missing *faminc*, simply drop them. As long as missingness is unrelated to factors affecting y , the remaining sample is still random.

The Missing Indicator Method (MIM)

Suppose only x_k is sometimes missing. **MIM creates two new variables:**

- ▶ z_{ik} : equals x_{ik} when observed, **0 otherwise**.
- ▶ m_{ik} : a "missing indicator" — 1 if missing, 0 if observed.

Missing Indicator Method

Run the regression

$$y_i \text{ on } x_{i1}, \dots, x_{i,k-1}, z_{ik}, m_{ik}, i = 1, \dots, n.$$

Use the **full sample** of n observations.

Appeal: Uses all n observations rather than just the complete cases.

Catch: Statistically valid **only under strong assumptions** — essentially requires x_k uncorrelated with other regressors (Jones 1996; Abrevaya & Donald 2017). These conditions **rarely hold** in practice.

Why MIM Needs a Strong Assumption

Zero-imputing x_k makes the fitted model differ across the two groups:

- ▶ **observed** ($m_k = 0$): $y = \beta_0 + \sum_{j < k} \beta_j x_j + \beta_k x_k + u$ (the true model);
- ▶ **missing** ($m_k = 1$): $y = (\beta_0 + \gamma) + \sum_{j < k} \beta_j x_j + u$ ($\beta_k x_k$ dropped, replaced by the dummy γ).

For the missing rows MIM **throws away** $\beta_k x_k$ and lets the single dummy γ stand in for it. A dummy can absorb only a **level shift** — the *average* of the dropped term, $\gamma \approx \beta_k \mathbb{E}(x_k \mid \text{missing})$.

This works **only if** $\mathbb{E}(x_k \mid x_1, \dots, x_{k-1}, \text{missing})$ does not move with the other regressors — i.e. x_k is **uncorrelated** with them. If x_k is correlated with $x_1, \dots, x_{k-1}$, the dropped $\beta_k x_k$ varies with those regressors and is mis-attributed to $\beta_1, \dots, \beta_{k-1}$ — omitted-variable bias the dummy cannot fix (it shifts the *intercept*, not the slopes). **Contrast:** complete-case OLS is consistent under MCAR *without* this condition — it just uses fewer observations.

Why MIM Needs a Strong Assumption: The Algebra

Use the identity $x_k = z_k + m_k x_k$ (observed $\Rightarrow m_k = 0$, $x_k = z_k$; missing $\Rightarrow z_k = 0$).
Substituting into the true model $y = \beta_0 + \sum_{j < k} \beta_j x_j + \beta_k x_k + u$ gives

$$y = \beta_0 + \sum_{j < k} \beta_j x_j + \beta_k z_k + \underbrace{\beta_k m_k x_k}_{\text{not a MIM regressor}} + u.$$

MIM regresses y on $1, x_1, \dots, x_{k-1}, z_k, m_k$: it includes m_k but *omits* the term $\beta_k m_k x_k$.

Model the missing-data conditional mean of x_k as linear:

$$\mathbb{E}(x_k \mid x_1, \dots, x_{k-1}, m_k = 1) = \delta_0 + \sum_{j < k} \delta_j x_j.$$

On the missing rows the omitted term is $\beta_k(\delta_0 + \sum_{j < k} \delta_j x_j)$: the dummy soaks up the constant $\gamma = \beta_k \delta_0$, but each $\beta_k \delta_j x_j$ is collinear with the regressor x_j .

Hence $\text{plim } \hat{\beta}_j^{\text{MIM}} = \beta_j + (\text{a term } \propto \beta_k \delta_j)$. The bias vanishes for *all* $j \iff \delta_1 = \dots = \delta_{k-1} = 0 \iff x_k$ is **uncorrelated** with $x_1, \dots, x_{k-1}$ in the missing subpopulation. Otherwise the slopes are biased.

MCAR vs. MIM: Which to Use?

Property	Complete Cases (MCAR)	Missing Indicator Method
Sample size used	Smaller	Full n
Required assumption	MCAR	MCAR + $x_k \perp$ other regressors
Robustness	More robust	Less robust
Bias if assumption fails	Inconsistent	Inconsistent

Wooldridge's recommendation: The complete cases estimator is generally preferred because it requires **weaker assumptions**. MIM should be used cautiously — it can cause more harm than good if its assumptions fail.

More advanced: Multiple imputation, EM algorithm — beyond scope.

Nonrandom Samples (§9-5b)

Even without missing data, the **sample itself** may be nonrandomly selected.

Two Types of Sample Selection

1. **Exogenous sample selection:** Selection based on the *independent* variables.
Example: We sample only people over 35 in a savings study.
2. **Endogenous sample selection:** Selection based on the *dependent* variable (or unobservables affecting y).
Example: We sample only households with wealth $< \$250,000$.

Statistical consequence:

- ▶ Exogenous selection $\Rightarrow$ OLS is **still unbiased and consistent**.
- ▶ Endogenous selection $\Rightarrow$ OLS is **biased and inconsistent**.

Sample Selection: Examples

Exogenous Selection: Saving Study

$$saving = \beta_0 + \beta_1 income + \beta_2 age + \beta_3 size + u.$$

Sample only people over 35. Selection is on *age* (a regressor). **OLS is unbiased.** Just a smaller, less general sample.

Endogenous Selection: Wealth Equation

$$wealth = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 age + u.$$

Sample only households with *wealth* < \$250,000. Selection is on *y*. **OLS is biased.** The conditional expectation in the truncated population is not the same as in the full population.

Stratified sampling: If sampling weights vary by strata defined by **exogenous** variables, OLS on the unweighted data is consistent. If strata are defined by endogenous variables, special methods are needed.

Wage Equations and Working Samples

A subtle case: estimating wage equations using **only working individuals**.

$$\log(\text{wage}^o) = \beta_0 + \beta_1 \text{educ} + \beta_2 \text{exper} + u, \quad (9.39)$$

where wage^o = wage *offer* (wage someone could earn if they worked).

- ▶ For non-workers, wage^o is unobserved.
- ▶ We estimate using only those who chose to work.
- ▶ If the choice to work depends on **unobserved factors** affecting wage offers (motivation, family circumstances) $\Rightarrow$ **endogenous selection**.

Heckman's solution: The Heckman two-step procedure (Ch. 17) corrects for sample selection bias when the selection mechanism can be modeled.

Outliers and Influential Observations (§9-5c)

Outlier vs. Influential Observation

- ▶ **Outlier:** An observation with unusual y and/or x values relative to the rest of the sample.
- ▶ **Influential observation:** An outlier whose removal substantially *changes* the OLS estimates.

Why OLS is sensitive: OLS minimizes the *sum of squared* residuals. Large residuals are weighted heavily (u^2 grows quadratically), so single observations can dominate the fit.

Two sources of outliers:

1. **Data entry errors** — always check summary statistics (min, max).
2. **Genuine extreme cases** — small populations, mixtures of distributions.

Example 9.8: R&D Intensity and Firm Size (RDCHEM)

Data: 32 chemical companies, $rdintens$ = R&D as % of sales, $sales$ in millions.

All 32 firms:

$$\widehat{rdintens} = 2.625 + 0.000053 \text{ sales} + 0.0446 \text{ profmarg},$$

with standard errors 0.586, 0.000044, 0.0462. $R^2 = 0.076$. **No coefficient is significant at 10%.**

Drop largest firm ($sales \approx \$40$ billion, twice the next largest):

$$\widehat{rdintens} = 2.297 + 0.000186 \text{ sales} + 0.0478 \text{ profmarg},$$

with new standard errors 0.592, 0.000084, 0.0445. $R^2 = 0.173$.

Striking result: Coefficient on $sales$ **more than triples**, becomes **significant at 5%**. The single large firm was masking the relationship.

Detecting Outliers: Studentized Residuals

Are large residuals a reliable sign of outliers? **Not directly.**

Why raw residuals mislead: OLS *fits* to outliers, so the residual at an outlier may actually be **small** (the line is dragged toward the outlier).

Studentized Residuals

The **studentized residual** for observation h is the OLS residual divided by an estimate of its standard deviation, computed **leaving observation h out**.

Easy computation trick: Add a dummy variable d_h for observation h to the full regression. The **t -statistic** on d_h is the studentized residual for h , with a t_{n-k-2} distribution.

Decision rule: A large absolute studentized residual ($|t| > 2$ or 3) flags observation h as an outlier.

Studentized Residuals: Caveats

In **Example 9.8** (32 firms):

- ▶ Largest firm's studentized residual: -1.82 ($p = 0.08$). Marginally significant but not extreme.
- ▶ First observation ($rdintens \approx 9.42$) has studentized residual 4.56 — much larger.

Surprise: Dropping the first observation *barely* changes the estimates. The largest studentized residual is **not** the most influential observation.

Lesson: Studentized residuals identify *extreme residuals*, not necessarily *influential observations*.

Best practice:

- ▶ Plot the data — visualize before computing.
- ▶ Run regression **with and without** suspect observations.
- ▶ Report differences honestly; let readers judge.
- ▶ **Do not just delete** unless the observation is clearly a data entry error.

Example 9.10: Infant Mortality (INFMRT)

$n = 51$ states + DC. All variables in logs:

$$\widehat{infmort} = 33.86 - 4.68 \log(pcinc) + 4.15 \log(physic) - 0.088 \log(popul).$$

Standard errors: 20.4, 2.6, 1.5, 0.29. $R^2 = 0.139$.

Surprise: more physicians per capita $\Rightarrow$ *higher* infant mortality?

DC is unusual: highest infant mortality (20.7 vs. 12.4 max elsewhere) AND most physicians (615 vs. 337 max elsewhere). Both extreme.

Drop DC:

$$\widehat{infmort} = 23.95 - 0.57 \log(pcinc) - 2.74 \log(physic) + 0.629 \log(popul).$$

Now $\log(physic)$ has the **expected negative sign** and is significant.

Conclusion: DC was **driving** the wrong sign. Reasonable to exclude (DC has unique demographic concentrations).

Functional Form and Outlier Sensitivity

Some functional forms are **less sensitive** to outliers.

Logarithmic transformations narrow the range of economic variables and reduce outlier influence.

Example 9.9 (RDCHEM in logs): Re-estimate the R&D model in log–log:

$$\log(rd) = \beta_0 + \beta_1 \log(sales) + \beta_2 profmarg + u.$$

All 32 firms:

$$\widehat{\log(rd)} = -4.378 + 1.084 \log(sales) + 0.0217 profmarg, R^2 = 0.918.$$

Drop largest firm:

$$\widehat{\log(rd)} = -4.404 + 1.088 \log(sales) + 0.0218 profmarg, R^2 = 0.904.$$

Practically identical results. The log–log specification is much more robust to outliers — a strong argument for using logs in skewed economic data.

Least Absolute Deviations Estimation (§9-6)

Least Absolute Deviations: Motivation & Setup

Motivation. OLS minimizes *squared* residuals, so a single extreme y can swing every coefficient. Rather than hunt down and delete outliers, can we estimate with a rule that is **intrinsically resistant** to them?

Setup. Replace OLS's quadratic loss with an **absolute-value** loss — minimize the sum of *absolute* residuals rather than squared ones. This estimates the **conditional median** instead of the mean. **Plan:** the LAD criterion, why the median is robust to outliers, and the robustness–efficiency trade-off against OLS.

Least Absolute Deviations (LAD)

Rather than identifying and removing outliers, an alternative: use an estimator **less sensitive** to outliers.

LAD Estimator

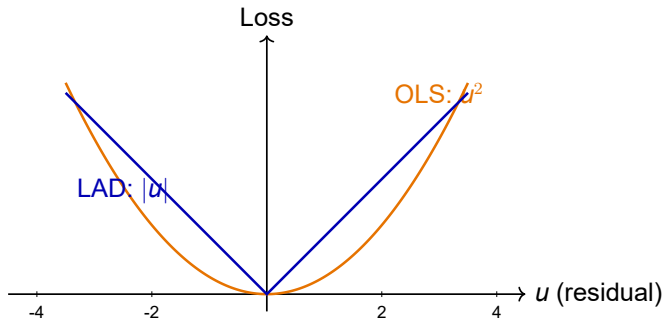
The LAD estimators of $\beta_0, \beta_1, \dots, \beta_k$ **minimize the sum of absolute residuals**:

$$\min_{b_0, b_1, \dots, b_k} \sum_{i=1}^n |y_i - b_0 - b_1 x_{i1} - \dots - b_k x_{ik}|. \quad (9.45)$$

Key contrast with OLS:

- ▶ **OLS** minimizes $\sum (\text{residual})^2$ — large residuals heavily penalized.
- ▶ **LAD** minimizes $\sum |\text{residual}|$ — outliers receive **linear** penalty, not quadratic.

LAD vs. OLS: Visual Comparison



Reading the picture:

- ▶ OLS loss grows **quadratically** — large residuals dominate.
- ▶ LAD loss grows **linearly** — large residuals get bounded weight.
- ▶ LAD is **robust to outliers** in y .

LAD Estimates the Conditional Median

LAD and the Conditional Median

OLS estimates the **conditional mean**: $\mathbb{E}(y \mid x_1, \dots, x_k)$.

LAD estimates the **conditional median**: $\text{Med}(y \mid x_1, \dots, x_k)$.

Why this matters: The median is not affected by extreme observations in the same way as the mean. So LAD estimates are **resilient** to outliers in y .

Symmetry caveat: If the conditional distribution of y given x 's is **symmetric**, then **mean = median**, and LAD and OLS estimate the same parameters.

If asymmetric: LAD and OLS estimate *different* quantities. The choice depends on **whether you care about the mean or the median** effect.

LAD: Computation and Inference

Computation:

- ▶ No closed-form solution (unlike OLS).
- ▶ Requires **linear programming**.
- ▶ Historically computationally heavy; modern computers handle large samples easily.

Inference:

- ▶ No **exact** small-sample distributions (t , F).
- ▶ Statistical inference is **asymptotic** (large-sample).
- ▶ Software reports asymptotic standard errors based on the conditional density of the error at zero.
- ▶ Formulas are more complex than OLS — see Koenker (2005) for treatment.

Practical note: For modest samples ($n < 200$) the asymptotic approximation may be poor — bootstrap is often preferred.

When to Choose LAD over OLS

Scenario	Use OLS	Use LAD
No outliers, symmetric errors	✓ (more efficient)	—
Outliers in y	risky	✓ (robust)
Asymmetric errors	estimates mean	estimates median
Skewed dependent variable	risky	✓ (often more interpretable)
Need exact small-sample inference	✓	×
Large samples	✓	✓
Mean is the policy quantity	✓	×
Median is the policy quantity	×	✓

Recommendation: Always run **both OLS and LAD** as a robustness check on important results. Differences between them *point to specific problems* (outliers, asymmetry).

Quantile Regression Generalization

LAD estimates the **conditional median**. We can generalize to other quantiles.

Quantile Regression (Koenker–Bassett 1978)

For quantile $\tau \in (0, 1)$, the τ -th quantile regression minimizes:

$$\min_{b_0, \dots, b_k} \sum_{i=1}^n \rho_{\tau}(y_i - b_0 - b_1 x_{i1} - \dots - b_k x_{ik}),$$

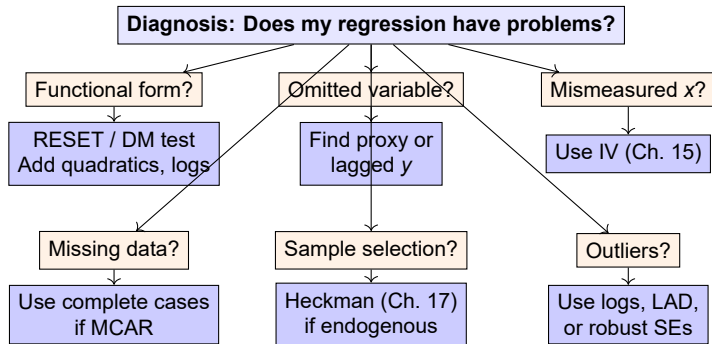
where $\rho_{\tau}(u) = u(\tau - \mathbf{1}\{u < 0\})$ is the **check function**.

Special case: $\tau = 0.5$ gives LAD (median regression).

Why useful:

- ▶ Different effect at the **bottom** ($\tau = 0.1$) vs. **top** ($\tau = 0.9$) of the conditional distribution of y .
- ▶ E.g., effect of education on *low-wage* workers vs. *high-wage* workers can be very different.

Practical Decision Tree



Chapter 9 Summary

Specification problems and solutions:

- ▶ **Functional form:** RESET, Davidson–MacKinnon, add quadratics/logs.
- ▶ **Omitted variables:** Proxy variables, lagged dependent variables.
- ▶ **Random slopes:** OLS works (with robust SEs) for the average effect.
- ▶ **Measurement error in y :** OLS unbiased; just larger SEs.
- ▶ **Measurement error in x (CEV):** attenuation bias toward zero.

Data problems:

- ▶ **Missing data (MCAR):** complete cases works; MIM is risky.
- ▶ **Exogenous selection:** OLS unbiased.
- ▶ **Endogenous selection:** OLS biased — needs Heckman or similar.
- ▶ **Outliers:** Detect with studentized residuals; use logs or LAD for robustness.

Looking Ahead

Limitation of Chapter 9: All solutions here use **OLS on a single cross-section**. Some problems (especially measurement error in x , endogenous selection) cannot be fully fixed this way.

Tools we will add later:

- ▶ **Time series** (Chs. 10–12): repeated observations over time.
- ▶ **Panel data** (Chs. 13–14): cross-section + time series.
- ▶ **Instrumental variables / 2SLS** (Ch. 15): direct fix for endogeneity.
- ▶ **Limited dependent variables** (Ch. 17): selection corrections, censored data.

Thank you!

Next: Chapter 10 (Time Series Basics)